

8 Integration Techniques and Improper Integrals



8.6

Numerical Integration

Objectives

- Approximate a definite integral using the Trapezoidal Rule.
- Approximate a definite integral using Simpson's Rule.
- Analyze the approximate errors in the Trapezoidal Rule and Simpson's Rule.



The Trapezoidal Rule

The Trapezoidal Rule

Some elementary functions simply do not have antiderivatives that are elementary functions. For example, there is no elementary function that has any of the following functions as its derivative.

$$\sqrt[3]{x}\sqrt{1-x}, \quad \sqrt{x}\cos x, \quad \frac{\cos x}{x}, \quad \sqrt{1-x^3}, \quad \sin x^2$$

If you need to evaluate a definite integral involving a function whose antiderivative cannot be found, then while the Fundamental Theorem of Calculus is still true, it cannot be easily applied. In this case, it is easier to resort to an approximation technique.

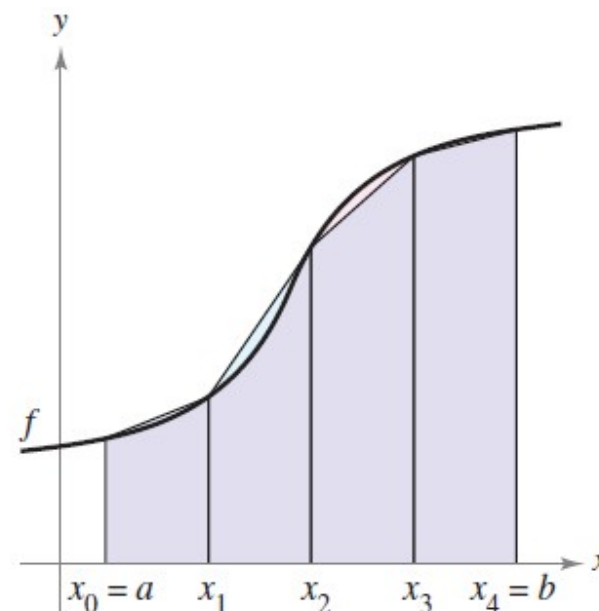
The Trapezoidal Rule

One way to approximate a definite integral is to use n trapezoids, as shown in Figure 8.14.

In the development of this method, assume that f is continuous and positive on the interval $[a, b]$. So, the definite integral

$$\int_a^b f(x) \, dx$$

represents the area of the region bounded by the graph of f and the x -axis, from $x = a$ to $x = b$.



The area of the region can be approximated using four trapezoids.

Figure 8.14

The Trapezoidal Rule

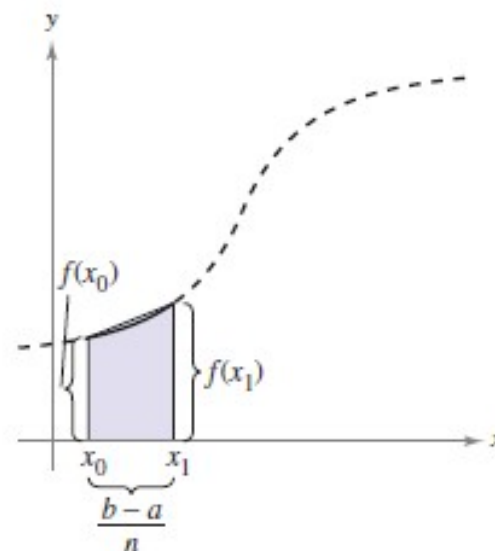
First, partition the interval $[a, b]$ into n subintervals, each of width $\Delta x = (b - a)/n$, such that

$$a = x_0 < x_1 < x_2 < \cdots < x_n = b.$$

Then form a trapezoid for each subinterval (see Figure 8.15).

The area of the i th trapezoid is

$$\text{Area of } i\text{th trapezoid} = \left[\frac{f(x_{i-1}) + f(x_i)}{2} \right] \left(\frac{b - a}{n} \right).$$



The area of the first trapezoid is

$$\left[\frac{f(x_0) + f(x_1)}{2} \right] \left(\frac{b - a}{n} \right).$$

Figure 8.15

The Trapezoidal Rule

This implies that the sum of the areas of the n trapezoids is

$$\begin{aligned}\text{Area} &= \left(\frac{b-a}{n}\right) \left[\frac{f(x_0) + f(x_1)}{2} + \dots + \frac{f(x_{n-1}) + f(x_n)}{2} \right] \\ &= \left(\frac{b-a}{2n}\right) [f(x_0) + f(x_1) + f(x_1) + f(x_2) + \dots + f(x_{n-1}) + f(x_n)] \\ &= \left(\frac{b-a}{2n}\right) [f(x_0) + 2f(x_1) + 2f(x_2) + \dots + 2f(x_{n-1}) + f(x_n)].\end{aligned}$$

The Trapezoidal Rule

Letting $\Delta x = (b - a)/n$, you can take the limit as $n \rightarrow \infty$ to obtain

$$\begin{aligned} & \lim_{n \rightarrow \infty} \left(\frac{b - a}{2n} \right) [f(x_0) + 2f(x_1) + \cdots + 2f(x_{n-1}) + f(x_n)] \\ &= \lim_{n \rightarrow \infty} \left[\frac{[f(a) - f(b)] \Delta x}{2} + \sum_{i=1}^n f(x_i) \Delta x \right] \\ &= \lim_{n \rightarrow \infty} \frac{[f(a) - f(b)](b - a)}{2n} + \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i) \Delta x \\ &= 0 + \int_a^b f(x) dx. \end{aligned}$$

The result is summarized in the next theorem.

The Trapezoidal Rule

THEOREM 8.3 The Trapezoidal Rule

Let f be continuous on $[a, b]$. The Trapezoidal Rule for approximating $\int_a^b f(x) dx$ is

$$\int_a^b f(x) dx \approx \frac{b-a}{2n} [f(x_0) + 2f(x_1) + 2f(x_2) + \cdots + 2f(x_{n-1}) + f(x_n)].$$

Moreover, as $n \rightarrow \infty$, the right-hand side approaches $\int_a^b f(x) dx$.

Example 1 – Approximation with the Trapezoidal Rule

Use the Trapezoidal Rule to approximate

$$\int_0^{\pi} \sin x \, dx.$$

Compare the results for $n = 4$ and $n = 8$, as shown in Figure 8.16.

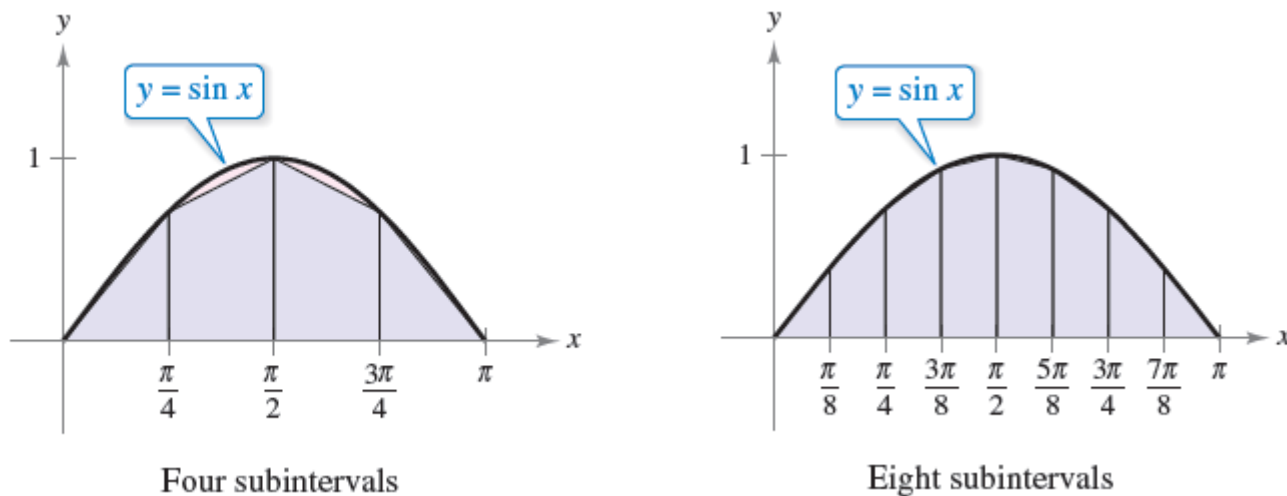


Figure 8.16

Example 1 – *Solution*

When $n = 4$, $\Delta x = \pi/4$, and you obtain

$$\begin{aligned}\int_0^{\pi} \sin x \, dx &\approx \frac{\pi}{8} \left(\sin 0 + 2 \sin \frac{\pi}{4} + 2 \sin \frac{\pi}{2} + 2 \sin \frac{3\pi}{4} + \sin \pi \right) \\ &= \frac{\pi}{8} (0 + \sqrt{2} + 2 + \sqrt{2} + 0) \\ &= \frac{\pi(1 + \sqrt{2})}{4} \\ &\approx 1.896.\end{aligned}$$

Example 1 – *Solution*

cont'd

When $n = 8$, $\Delta x = \pi/8$, and you obtain

$$\begin{aligned}\int_0^{\pi} \sin x \, dx &\approx \frac{\pi}{16} \left(\sin 0 + 2 \sin \frac{\pi}{8} + 2 \sin \frac{\pi}{4} + 2 \sin \frac{3\pi}{8} + 2 \sin \frac{\pi}{2} \right. \\ &\quad \left. + 2 \sin \frac{5\pi}{8} + 2 \sin \frac{3\pi}{4} + 2 \sin \frac{7\pi}{8} + \sin \pi \right) \\ &= \frac{\pi}{16} \left(2 + 2\sqrt{2} + 4 \sin \frac{\pi}{8} + 4 \sin \frac{3\pi}{8} \right) \\ &\approx 1.974.\end{aligned}$$

For this particular integral, you could have found an antiderivative and determined that the exact area of the region is 2.



Simpson's Rule

Simpson's Rule

One way to view the trapezoidal approximation of a definite integral is to say that on each subinterval, you approximate f by a *first*-degree polynomial. In Simpson's Rule, you take this procedure one step further and approximate f by *second*-degree polynomials.

Before presenting Simpson's Rule, consider the following theorem for evaluating integrals of polynomials of degree 2 (or less).

THEOREM 8.4 Integral of $p(x) = Ax^2 + Bx + C$

If $p(x) = Ax^2 + Bx + C$, then

$$\int_a^b p(x) dx = \left(\frac{b-a}{6}\right) \left[p(a) + 4p\left(\frac{a+b}{2}\right) + p(b) \right].$$

Simpson's Rule

To develop Simpson's Rule for approximating a definite integral, you again partition the interval $[a, b]$ into n subintervals, each of width $\Delta x = (b - a)/n$. This time, however, n is required to be even, and the subintervals are grouped in pairs such that

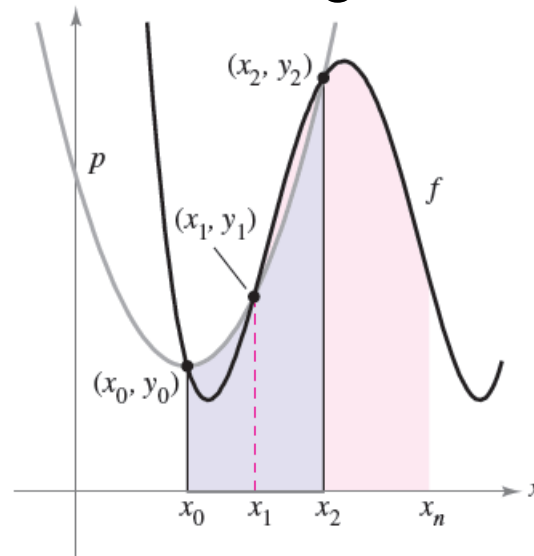
$$a = x_0 < x_1 < x_2 < x_3 < x_4 < \cdots < x_{n-2} < x_{n-1} < x_n = b.$$

$[x_0, x_2]$ $[x_2, x_4]$ $[x_{n-2}, x_n]$

On each (double) subinterval $[x_{i-2}, x_i]$, you can approximate f by a polynomial p of degree less than or equal to 2.

Simpson's Rule

For example, on the subinterval $[x_0, x_2]$, choose the polynomial of least degree passing through the points (x_0, y_0) , (x_1, y_1) , and (x_2, y_2) , as shown in Figure 8.17.



$$\int_{x_0}^{x_2} p(x) dx \approx \int_{x_0}^{x_2} f(x) dx$$

Figure 8.17

Simpson's Rule

Now, using p as an approximation of f on this subinterval, you have, by Theorem 8.4,

$$\begin{aligned}\int_{x_0}^{x_2} f(x) \, dx &\approx \int_{x_0}^{x_2} p(x) \, dx \\&= \frac{x_2 - x_0}{6} \left[p(x_0) + 4p\left(\frac{x_0 + x_2}{2}\right) + p(x_2) \right] \\&= \frac{2[(b - a)/n]}{6} [p(x_0) + 4p(x_1) + p(x_2)] \\&= \frac{b - a}{3n} [f(x_0) + 4f(x_1) + f(x_2)].\end{aligned}$$

Repeating this procedure on the entire interval $[a, b]$ produces the next theorem.

Simpson's Rule

THEOREM 8.5 Simpson's Rule

Let f be continuous on $[a, b]$ and let n be an even integer. Simpson's Rule for approximating $\int_a^b f(x) dx$ is

$$\int_a^b f(x) dx \approx \frac{b-a}{3n} [f(x_0) + 4f(x_1) + 2f(x_2) + 4f(x_3) + \cdots + 2f(x_{n-2}) + 4f(x_{n-1}) + f(x_n)].$$

Moreover, as $n \rightarrow \infty$, the right-hand side approaches $\int_a^b f(x) dx$.

Example 2 – *Approximation with Simpson's Rule*

Use Simpson's Rule to approximate

$$\int_0^{\pi} \sin x \, dx.$$

Compare the results for $n = 4$ and $n = 8$.

Solution:

When $n = 4$, you have

$$\int_0^{\pi} \sin x \, dx \approx \frac{\pi}{12} \left(\sin 0 + 4 \sin \frac{\pi}{4} + 2 \sin \frac{\pi}{2} + 4 \sin \frac{3\pi}{4} + \sin \pi \right) \approx 2.005.$$

When $n = 8$, you have $\int_0^{\pi} \sin x \, dx \approx 2.0003$.



Error Analysis

Error Analysis

When you use an approximation technique, it is important to know how accurate you can expect the approximation to be. The following theorem, gives the formulas for estimating the errors involved in the use of Simpson's Rule and the Trapezoidal Rule.

THEOREM 8.6 Errors in the Trapezoidal Rule and Simpson's Rule

If f has a continuous second derivative on $[a, b]$, then the error E in approximating $\int_a^b f(x) dx$ by the Trapezoidal Rule is

$$|E| \leq \frac{(b-a)^3}{12n^2} [\max |f''(x)|], \quad a \leq x \leq b. \quad \text{Trapezoidal Rule}$$

Moreover, if f has a continuous fourth derivative on $[a, b]$, then the error E in approximating $\int_a^b f(x) dx$ by Simpson's Rule is

$$|E| \leq \frac{(b-a)^5}{180n^4} [\max |f^{(4)}(x)|], \quad a \leq x \leq b. \quad \text{Simpson's Rule}$$

Error Analysis

In general, when using an approximation, you can think of the error E as the difference between $\int_a^b f(x) dx$ and the approximation.

Theorem 8.6 states that the errors generated by the Trapezoidal Rule and Simpson's Rule have upper bounds dependent on the extreme values of $f''(x)$ and $f^{(4)}(x)$ in the interval $[a, b]$. Furthermore, these errors can be made arbitrarily small by *increasing* n , provided that f'' and $f^{(4)}$ are continuous and therefore bounded in $[a, b]$.

Example 3 – *The Approximate Error in the Trapezoidal Rule*

Determine a value of n such that the Trapezoidal Rule will approximate the value of

$$\int_0^1 \sqrt{1 + x^2} \, dx$$

with an error that is less than or equal to 0.01.

Solution:

Begin by letting $f(x) = \sqrt{1 + x^2}$ and finding the second derivative of f .

$$f'(x) = x(1 + x^2)^{-1/2} \quad \text{and} \quad f''(x) = (1 + x^2)^{-3/2}$$

The maximum value of $|f''(x)|$ on the interval $[0, 1]$ is $|f''(0)| = 1$.

Example 3 – *Solution*

cont'd

So, by Theorem 8.6, you can write

$$\begin{aligned}|E| &\leq \frac{(b-a)^3}{12n^2} |f''(0)| = \frac{1}{12n^2}(1) \\ &= \frac{1}{12n^2}.\end{aligned}$$

To obtain an error E that is less than or equal to 0.01, you must choose n such that $1/(12n^2) \leq 1/100$.

$$100 \leq 12n^2 \quad \Rightarrow \quad n \geq \sqrt{\frac{100}{12}} \approx 2.89$$

Example 3 – *Solution*

cont'd

So, you can choose $n = 3$ (because n must be greater than or equal to 2.89) and apply the Trapezoidal Rule, as shown in Figure 8.18,

to obtain

$$\begin{aligned}\int_0^1 \sqrt{1+x^2} dx &\approx \frac{1}{6} \left[\sqrt{1+0^2} + 2\sqrt{1+\left(\frac{1}{3}\right)^2} \right. \\ &\quad \left. + 2\sqrt{1+\left(\frac{2}{3}\right)^2} + \sqrt{1+1^2} \right] \\ &\approx 1.154.\end{aligned}$$

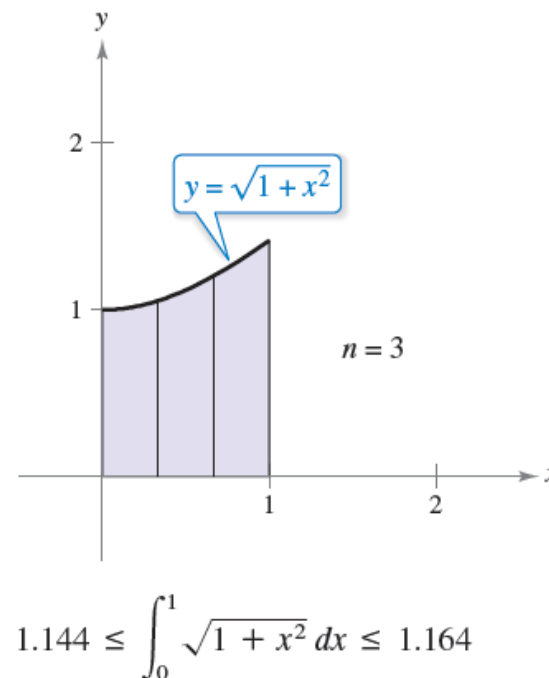


Figure 8.18

Example 3 – *Solution*

cont'd

So, by adding and subtracting the error from this estimate, you know that,

$$1.144 \leq \int_0^1 \sqrt{1+x^2} \, dx \leq 1.164.$$